

Time: 1 Hrs.

Total Marks: 40

Note: Marks are indicated against each question.

Question No. 1

(8)

If P, Q and R are three atomic variables, obtain the Principal Disjunctive Normal form and Principal Conjunctive Normal form for the given formula without using TT.

$$(P \rightarrow (Q \wedge R)) \vee (\sim P \rightarrow (Q \vee R))$$

Question no. 2

(8)

Determine if the following inference pattern is Valid OR Invalid. Produce evidence (write the steps) to confirm its validity/invalidity.

$$\frac{(p \vee q) \rightarrow (s \wedge t) \quad [\sim((\sim s) \vee (\sim t))] \rightarrow [(\sim r) \vee q]}{(p \vee q) \rightarrow (r \rightarrow q)}$$

Question no. 3

(2+6)

What is a partition on a set? Let $A = \{1, 2, 3, 4, 5\}$ and a relation R defined on A is $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$. Obtain the cells of the partition.

Question no. 4

(1+3+4)

What is a Hasse diagram? Draw the Hasse diagrams of the following sets under the partial ordering relation "divides" and indicate those which are totally ordered. Also, determine least member, greatest member, minimal member and maximal member in each case.

- (i) $\{2, 6, 24\}$
- (ii) $\{1, 2, 3, 6, 12\}$
- (iii) $\{3, 9, 27, 54\}$

Question no. 5 What is a lattice? Determine (show steps), which of the following graphs are lattice? (2+6)

